

## Algebra First-Year Practice Exam A, Part 1

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*Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.*

1. Let  $G$  be a group of order 44. Prove that
  - (a) a Sylow 11-subgroup of  $G$  is normal in  $G$ , and
  - (b) the centre of  $G$  contains an element of order 2.
2. Let  $G = A \times B$  be a finite group which is the (internal) direct product of two subgroups  $A$  and  $B$  which are nonabelian simple groups.
  - (a) Show that the only proper, nontrivial normal subgroups of  $G$  are  $A$  and  $B$ . (Hint: If  $N$  is a proper normal subgroup different from  $A$  and  $B$ , consider the commutator  $[N, A]$ .)
  - (b) Give an example to show that (a) would be false were  $A$  and  $B$  allowed to be abelian.
3. Let  $G = \text{GL}_2(\mathbb{R})$  act on  $\mathbb{R}^2$  by matrix multiplication:  $A \cdot \mathbf{v} = A\mathbf{v}$  for  $A \in G$ ,  $\mathbf{v} \in \mathbb{R}^2$ .
  - (a) Determine the stabilizer in  $G$  of  $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{R}^2$ .
  - (b) Determine the orbit of  $\mathbf{e}_2$ .
  - (c) How many orbits are there for the action of  $G$  on  $\mathbb{R}^2$ ?
4. Let  $G$  be a group.
  - (a) Define the ascending central series
$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \dots$$
for  $G$ . (This is also known as the upper central series.)
  - (b) Prove that the groups  $Z_i(G)$  are all characteristic subgroups of  $G$ .
  - (c) Give an example of a nontrivial group  $G$  such that the groups  $Z_i(G)$  are all trivial.
5. Let  $n \geq 2$  and let  $Z_n$  be a cyclic group of order  $n$ . Prove that

$$\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$